

Carroll's Logic Game: Puzzling Out the Categorical Syllogism

An original lesson plan inspired by **Lewis Carroll's** *Symbolic Logic* (Project Gutenberg). Not affiliated with or endorsed by Project Gutenberg or Lewis Carroll. Explore the original course: <https://www.gutenberg.org/ebooks/28696>

GRADE LEVEL

Grades 9-12

DURATION

50 minutes

TOPICS

Logic

“Can working through a playful puzzle teach you to reason more rigorously than memorizing a rule ever could?”

Learning Objectives

By the end of this lesson, students will be able to:

- Translate whimsical premises, in the style of Lewis Carroll, into standard logical form and identify the shared middle term.
- Solve multi-premise 'sorites' style puzzles by chaining several syllogisms together to reach a single valid conclusion.
- Explain why Carroll's playful, absurd content is a deliberate pedagogical device for isolating logical form from real-world plausibility.

Materials Needed

- Handout with three or four of Carroll's original whimsical syllogism puzzles, adapted and simplified
- Blank sorites-chain worksheet with boxes for linking each premise's shared term to the next
- Whiteboard for building one puzzle chain together as a class
- A short excerpt handout describing who Lewis Carroll (Charles Dodgson) was and why he wrote logic puzzles for children and adults alike

Lesson Procedure

Hook: The Absurd Argument

6 min

Read aloud a Carroll-style premise set: 'No baby can manage a crocodile. No one who cannot manage a crocodile is despised.' Ask students to guess the conclusion before revealing it (babies are not despised), and ask why Carroll would deliberately choose such a ridiculous topic rather than something serious.

Direct Instruction: Why Silly Content Is the Point

10 min

Explain that Carroll, better known as the author of *Alice in Wonderland*, was a mathematician who used deliberately absurd, memorable content precisely so students couldn't rely on prior knowledge or gut feelings about the subject matter, forcing them to track the logical structure itself. Connect this to the categorical syllogism's shared middle terms, and introduce a 'sorites' as a chain of several linked syllogisms collapsing into one long argument with a single conclusion.

Guided Practice: Chain the Clues

15 min

As a whole class, work through one multi-premise Carroll puzzle together on the board, identifying which term repeats between which premises and crossing out the repeated middle terms one pair at a time until only the terms of the final conclusion remain.

Small-Group Challenge: Solve Your Own Sorites

14 min

In pairs, students receive a new, unsolved Carroll-style multi-premise puzzle and must use the same term-cancellation method to reach the valid conclusion, showing their work on the worksheet.

Closure: Share and Reflect

5 min

A few pairs share their solved conclusion and briefly explain their chain. Close by asking students to name one moment during the chain where they could have made an error if they weren't tracking the terms carefully.

Discussion Questions

- 1 Why might absurd, made-up content actually make logical structure easier to see than realistic content would?
- 2 What is the risk of using real-world, believable premises when practicing formal logic, compared to using deliberately silly ones?
- 3 In a long chain of premises, where are you most likely to make a mistake, and what habit could prevent it?
- 4 Is solving a sorites puzzle more like solving a math problem or more like reading a story? What does that suggest about the relationship between logic and other subjects?
- 5 Lewis Carroll wrote both these logic puzzles and *Alice in Wonderland's* nonsensical wordplay. What might connect his interest in logic to his interest in nonsense literature?

Assessment

Students solve two new Carroll-style sorites puzzles independently, showing the full chain of term-cancellations that leads to each valid conclusion, and write one sentence identifying which specific pair of terms let them cross out a shared middle term at each step. Grade for correct tracking of terms through the chain, not for whether the puzzle's content is realistic.

Extension Activity



Have students write their own three- or four-premise Carroll-style sorites puzzle using an absurd topic of their choosing, making sure each premise shares exactly one term with the next, and trade with a partner to solve. This requires working the logic backward from a valid conclusion, which is a stronger test of understanding than solving alone.

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